# MECHATRONICS AND MEMS

## Subject Code : 23A03702d

## UNIT I : Introduction

**Syllabus:** Definition of Mechatronics, Need for Mechatronics in Industry, Objectives of mechatronics, mechatronics design process, Mechatronics key elements, applications - CNC machines, Tool monitoring systems, FMS, Industrial Robots, Automatic packaging systems, Automatic inspection systems.

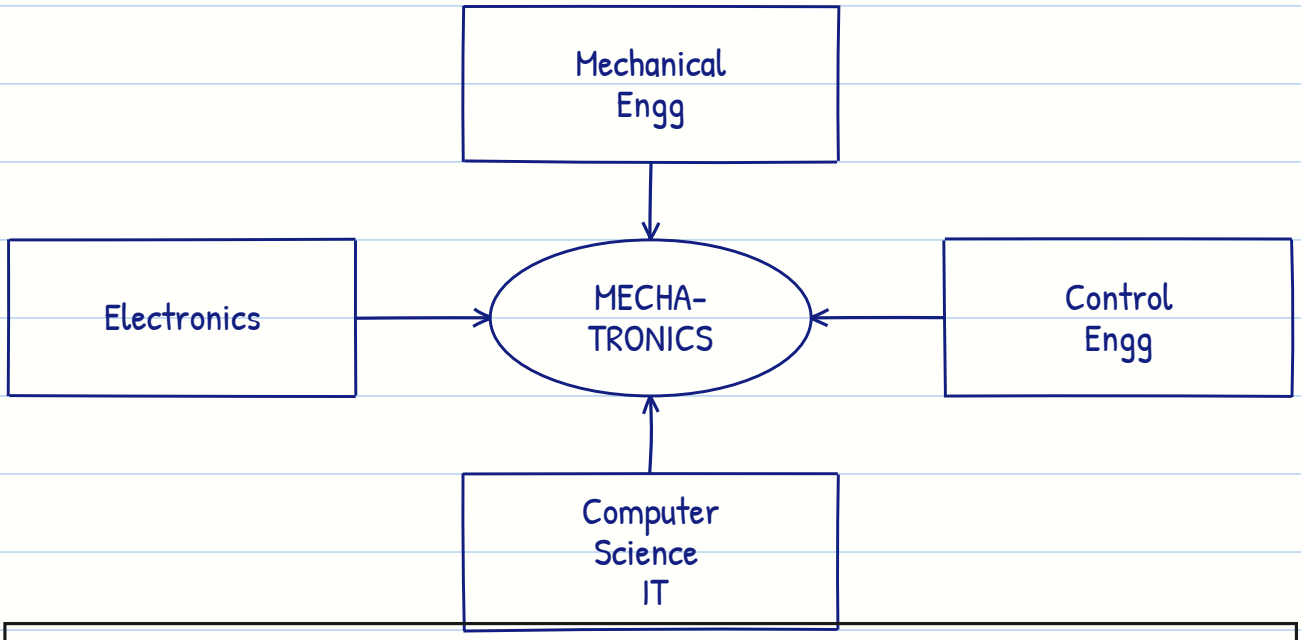
## I. Definition of Mechatronics

The word **Mechatronics** was coined in 1969 by **Tetsuro Mori** , an engineer at Yaskawa Electric Corporation, Japan. It is made from two parts: **MECHA** from Mechanics (mechanical engineering) and **TRONICS** from Electronics.

### Definitions

- Mechatronics is the **synergistic integration** of mechanical engineering, electronics, control engineering and computer science in the design and manufacture of products and processes.
- IEEE/ASME: "Synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of industrial products and processes."
- Mechatronics = Mechanical + Electronics + Electrical + Control + Computer (IT) working together as ONE system.

- A mechatronic system is **intelligent** : it senses the environment, processes the information and acts on it automatically.




Mechatronics = Mechanics + Electronics + Control + Computing

## Examples of mechatronic products

- Washing machine, microwave oven, digital camera, ATM, printers, CD/DVD drive, ABS braking and airbags in cars, CNC machine, robots, drones, 3D printer, automatic door, smart phone.

## Traditional vs Mechatronic approach

| Point | Traditional design | Mechatronic design |
|---|---|---|
| Approach | Sequential - mech first, then electrical | Concurrent - all domains together |
| Flexibility | Fixed by mechanical parts (cams, gears) | Changed by software |
| Accuracy | Limited, wear affects it | High, feedback corrects errors |
| Size & cost | Bulky, costly | Compact, cheaper |

| | | |
|---|---|---|
| Intelligence | None or little | Self-monitoring, decision making |

## 2. Need for Mechatronics in Industry

- **Global competition:** products must be cheap, good quality and reach the market fast (short time to market).
- **Customer demand:** customised products, small batch sizes, variety. Needs flexible machines.
- **High precision & repeatability:** micron-level accuracy which humans cannot give.
- **Productivity:** machines work 24 x 7 without fatigue, so output goes up and unit cost goes down.
- **Quality:** feedback sensors and automatic inspection reduce rejects (zero-defect aim).
- **Safety:** robots do dangerous jobs - painting, welding, handling hot or toxic materials.
- **Compact and light designs:** electronics replaces heavy mechanical linkages, cams and gear trains.
- **Reliability & less maintenance:** fewer moving parts, self-diagnosis and predictive maintenance.
- **Flexibility:** change of product only needs change of program - no new tooling.
- **Energy and material saving:** accurate control avoids wastage.

### Advantages

- Higher accuracy and speed, lower cost per piece, better quality.
- Flexible, programmable, easy to upgrade (software update).
- Remote monitoring and data collection (Industry 4.0, IoT).

### Disadvantages / limitations

- High initial investment and need for skilled multi-disciplinary engineers.
- Complex design, difficult fault finding (a fault may be in mechanics, electronics or software).
- Needs a stable power supply, and is affected by noise and vibration.

## 3. Objectives of Mechatronics

- To **integrate** mechanical, electronic, control and computer technologies for a better product.
- To build **smart / intelligent** products that adapt to changes in their surroundings.
- To increase **productivity, accuracy and repeatability** of machines.
- To reduce **cost, size, weight and energy consumption** of the product.
- To improve **reliability** and reduce maintenance.
- To get **flexibility** - the same hardware doing many tasks by changing software.
- To reduce **design time** and time-to-market by concurrent engineering.

- To improve **safety** and user friendliness (better man-machine interface).
- To enable **automation** of manufacturing and remote operation.

## 4. Mechatronics Design Process

Mechatronic design is **concurrent (simultaneous) engineering** : mechanical, electrical and software teams work together from the beginning, not one after another. It follows the **V-model** .

### Stages

- **Need / requirement analysis:** identify the customer need and write a clear specification (speed, load, accuracy, cost, size, safety).
- **Conceptual design:** generate alternative solutions, select the best with a decision matrix. Decide what is done in mechanics, in electronics and in software.
- **System-level design:** divide the system into subsystems (sensor, actuator, controller, mechanism) and fix interfaces between them.
- **Detailed design:** mechanical (CAD, stress), electrical (circuits, PCB), software (algorithms, code) and control design (PID tuning).
- **Modelling and simulation:** build mathematical models (MATLAB/Simulink) and check behaviour before making hardware.
- **Prototype & hardware-in-loop test:** integrate and test each module, then the whole system.
- **Validation and optimisation:** check against the specification, tune and improve.

- **Manufacture, documentation and maintenance.**

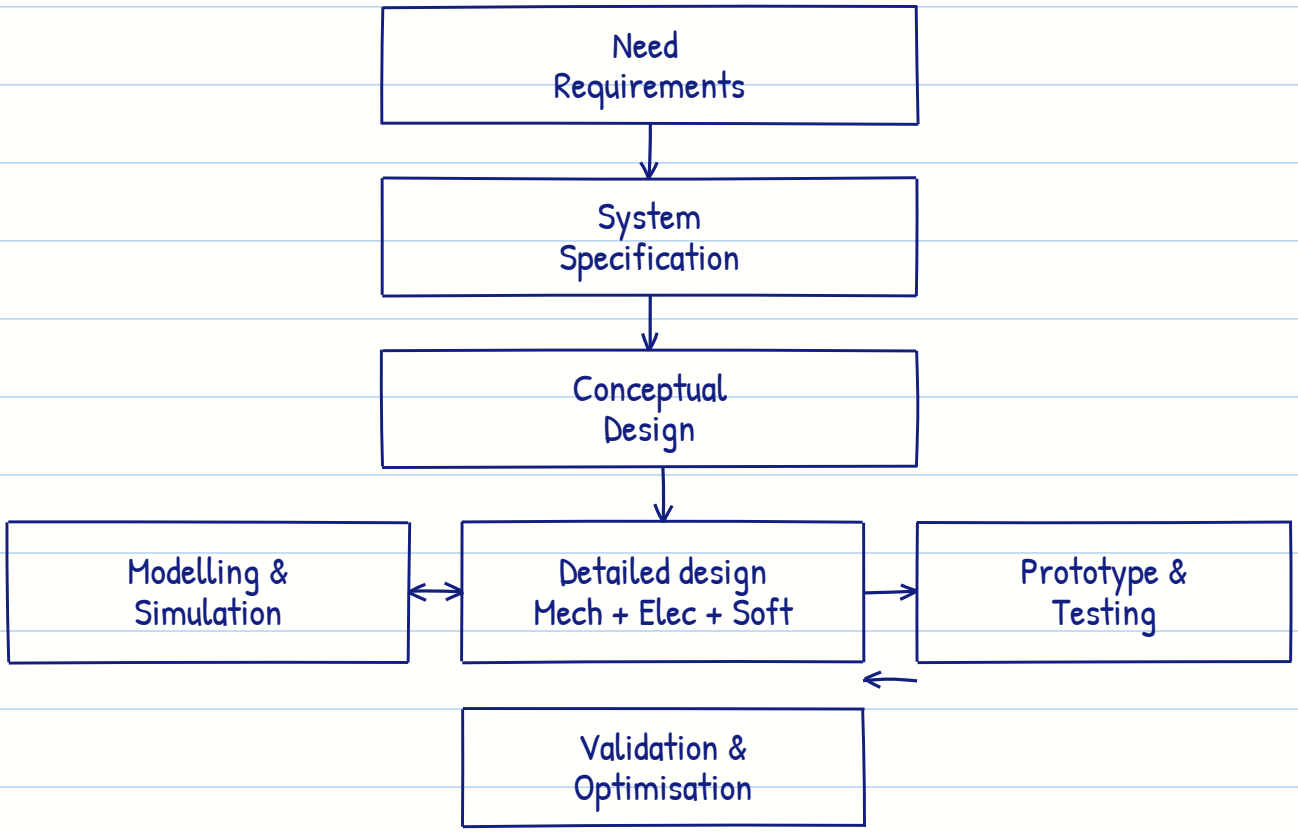



Fig: Mechatronic design process (V-model, concurrent engineering)

### Traditional vs Mechatronic design

- Traditional: design is sequential and "thrown over the wall" - errors are found late and costly.
- Mechatronic: parallel design, so the interactions between mechanics, electronics and software are checked early.

## 5. Key Elements of a Mechatronic System

A mechatronic system has **five key elements** :

- **1. Sensors:** convert a physical quantity (position, force, temperature, speed) into an electrical signal. Eg: encoder, LVDT, thermocouple, strain gauge.

- **2. Signal conditioning:** amplify, filter, convert (ADC/DAC), isolate and linearise the sensor signal so the controller can use it. Eg: op-amp, filters, ADC.
- **3. Controller (decision unit):** compares the actual value with the desired value and takes a decision. Eg: microprocessor, microcontroller, PLC, PC, DSP.
- **4. Actuators (with drivers):** convert the controller's electrical output into physical action - motion, force, heat. Eg: motors, solenoids, hydraulic and pneumatic cylinders, relays, piezo.
- **5. Mechanical system / Physical plant:** the mechanism that does the actual work. Eg: gears, linkages, slides, robot arm.
- Also: **Display / man-machine interface** (keypad, LCD, touch screen) and **Power supply** .

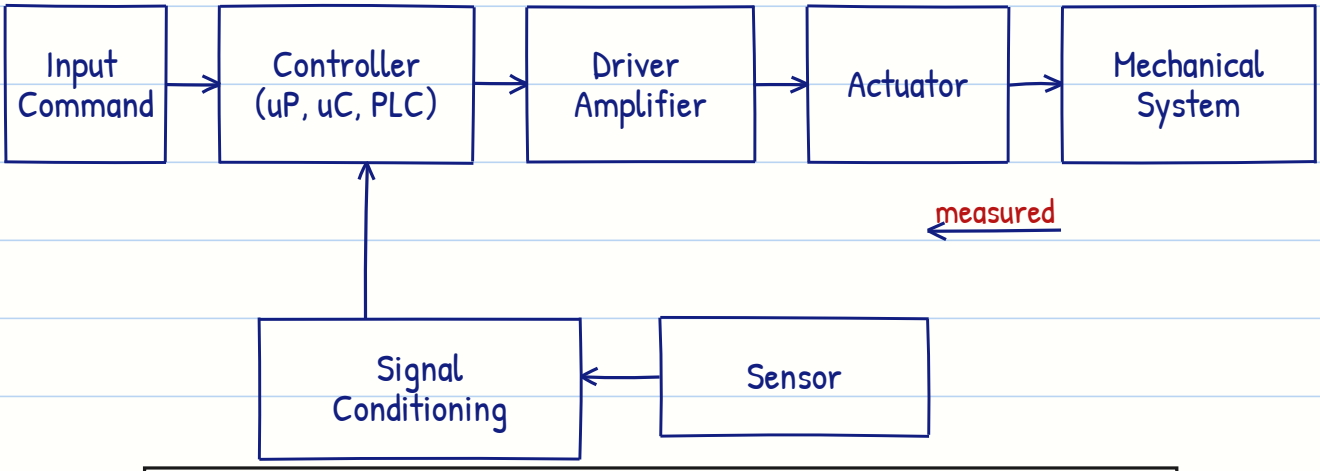



Fig: Closed-loop mechatronic system with feedback

### Control loops

- **Open loop:** no feedback; output has no effect on control action. Simple and cheap but not accurate. Eg: toaster, washing machine timer.
- **Closed loop:** the output is measured by a sensor and fed back to the controller to reduce error. Accurate. Eg: CNC servo axis, cruise control, robot joint.
- Error = Set point - Feedback value.

## 6. Applications of Mechatronics

Mechatronics is used in manufacturing, automobiles, aerospace, medicine, consumer goods, agriculture and defence. Industrial applications in the syllabus are explained below.

## 6.1 Computer Numerical Control (CNC) Machines

- A CNC machine is a machine tool whose motions (axes, spindle speed, feed, coolant) are controlled by a **computer reading a part program** (G and M codes).
- Eg: CNC lathe, milling machine, drilling machine, machining centre, EDM, laser cutter.

### Main parts

- **Part program / input device:** G-code written by the programmer (CAD/CAM) and sent through USB, LAN or keypad.
- **Machine Control Unit (MCU):** the computer. It has a Data

Processing Unit (reads and decodes the program) and a Control Loop Unit (sends signals to drives and takes feedback).

- **Drive system:** servo motor + ball screw for each axis; spindle motor.
- **Feedback devices:** encoders, resolvers and linear scales for position and velocity.
- **Machine tool:** bed, slides, spindle, tool turret/ATC, work holding.
- **Display & operator panel.**

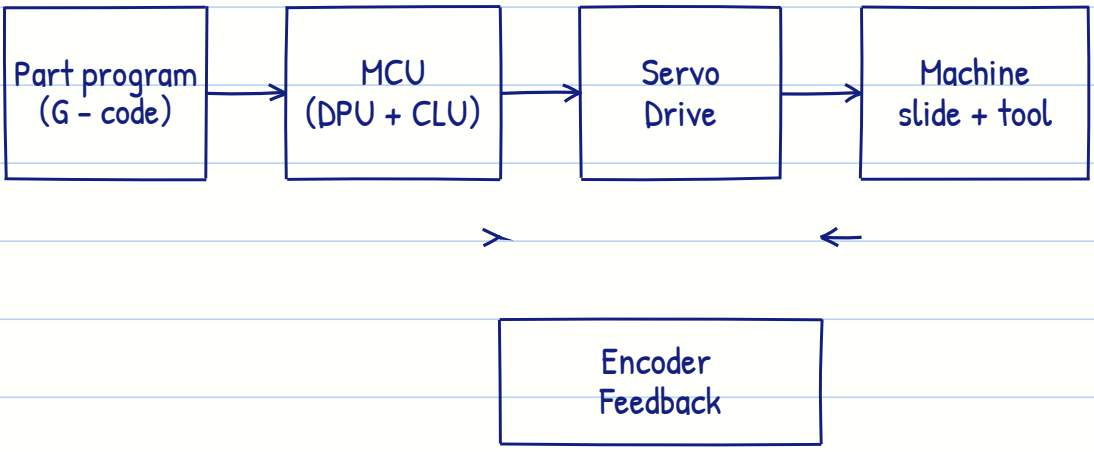



Fig: Block diagram of a CNC machine (closed loop)

## Types of control

- **Open loop:** stepper motor, no feedback. Cheap, less accurate.
- **Closed loop:** servo motor with feedback. High accuracy, used in most CNC machines.
- **Point-to-point** (drilling), **straight cut** and **contouring** (continuous path, 2 to 5 axes).

### Advantages

- High accuracy and repeatability, good surface finish, complex shapes possible.
- Less setup time and less operator skill, reduced scrap, can run unattended.

### Limitations

- Costly machine and maintenance, needs skilled programmer, and is not economic for very small jobs.

## 6.2 Tool Monitoring Systems

- Tool wear and breakage spoils the job, damages the machine and stops production. A **Tool Condition Monitoring (TCM)** system watches the tool continuously during cutting and gives a warning or stops the machine.
- Tool wear cannot be measured directly during cutting, so it is **indirectly** sensed.

### Methods

- **Direct methods:** optical/vision camera, laser, touch probe measure the tool dimension. Done between cuts.
- **Indirect methods:** measure a signal which changes with wear:
  - **Cutting force / torque** (dynamometer, strain gauge) - force rises as the tool wears.
  - **Spindle motor current / power** - increases with wear.
  - **Vibration** (accelerometer) and **acoustic emission (AE)** sensors -

sudden change shows breakage.

- **Temperature** of tool and chip (thermocouple, infrared).
- **Sound** (microphone) and surface roughness of the job.

**Working**

- Sensors -> signal conditioning (filter, amplifier, ADC) -> feature extraction -> **decision logic** (threshold, neural network, fuzzy) -> alarm or tool change command or machine stop.

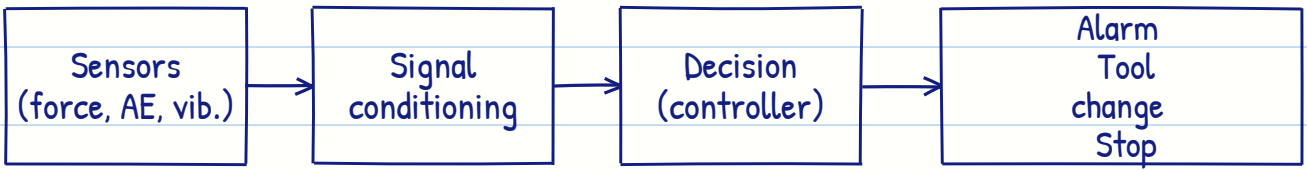



Fig: Tool condition monitoring system

- **Benefits:** avoids scrap and machine crash, increases tool life usage, allows unmanned (lights-out) machining, reduces cost.

## 6.3 Flexible Manufacturing System (FMS)

- FMS is an **automated manufacturing system** having a group of CNC machines connected by an automated material handling system, all controlled by a central computer, which can make a **variety of parts** with little changeover time.
- It lies between dedicated (transfer line) and stand-alone CNC machines - gives **medium volume, medium variety** .

## Main components

- **Workstations:** CNC machines (machining centres, lathes), inspection station (CMM), washing station.
- **Material handling system:** AGVs (automated guided vehicles), conveyors, robots, pallet changers, AS/RS (automated storage & retrieval system).
- **Central computer control:** schedules jobs, sends part programs (DNC), tracks tools and pallets, collects data.
- **Tool management:** tool magazines, tool presetting, automatic tool change.
- **Load/unload stations** and fixtures/pallets.

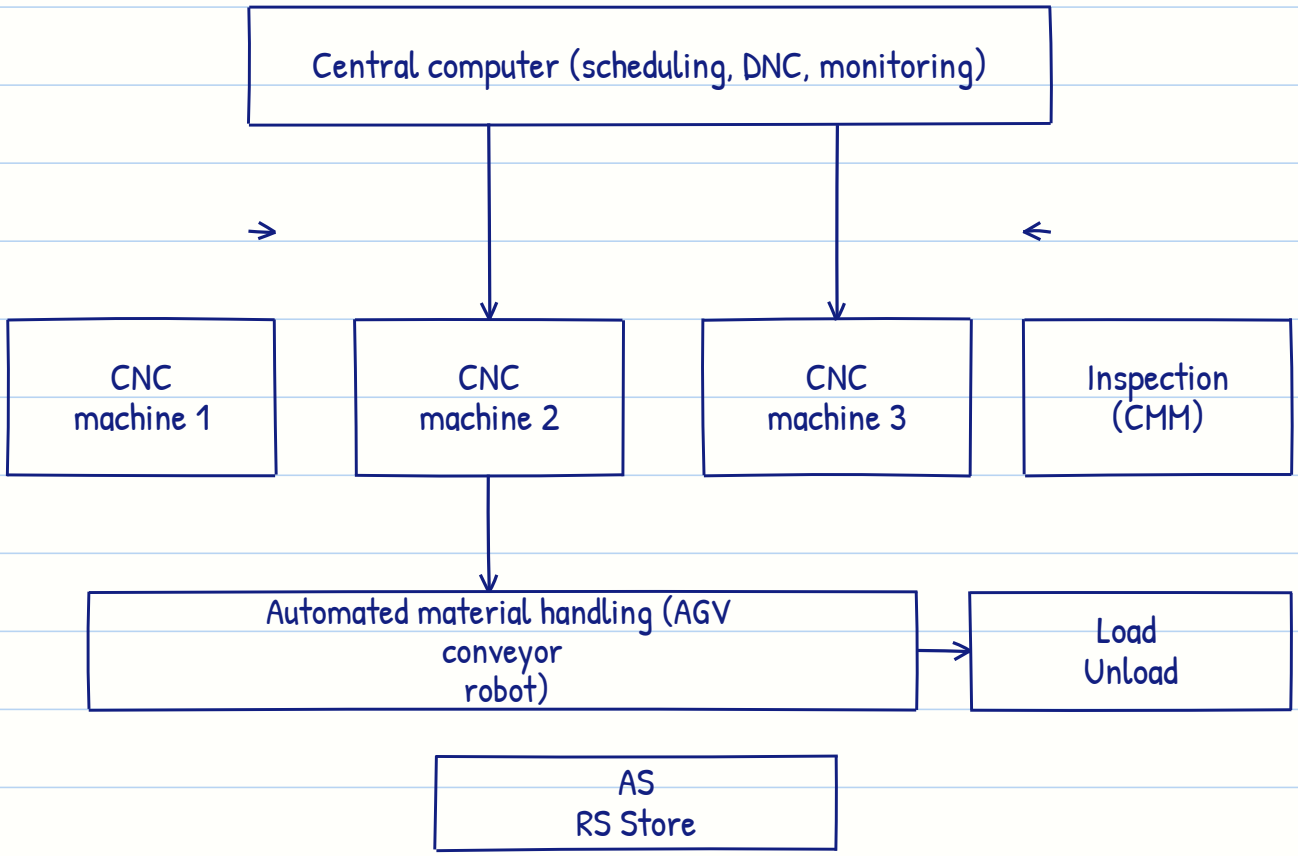



Fig: Layout of a flexible manufacturing system

### Types

- **Flexible manufacturing cell (FMC)** - 1 to 3 machines; **FMS** - 4 or more machines; **Flexible transfer line** - high volume.

### Advantages

- Reduced setup/changeover and lead time, high machine utilisation (up to 80-90%).
- Low work-in-process inventory, consistent quality, and quick response to design changes.

### Disadvantages

- Very high initial cost, complex software, needs skilled people, and the system stops if the central computer fails.

## 6.4 Industrial Robots

- **Definition (RIA):** A robot is a reprogrammable, multifunctional manipulator designed to move materials, parts, tools or special devices through variable programmed motions for performing a variety of tasks.

### Main parts of a robot

- **Manipulator (arm + wrist):** links and joints (revolute R, prismatic P). Gives the degrees of freedom (DOF).
- **End effector:** gripper, welding torch, spray gun, vacuum cup - the "hand" of the robot.
- **Actuators:** electric servo / stepper motors, hydraulic and pneumatic drives.

- **Sensors:** encoders, force/torque, vision, proximity - for feedback.
- **Controller:** the brain, runs program and does kinematics. Teach pendant for programming.
- **Power supply.**

### Classification (by configuration)

| Type | Joints | Work envelope / use |
|---|---|---|
| Cartesian | PPP | Rectangular - pick & place, CNC loading, 3D printing |
| Cylindrical | RPP | Cylindrical - assembly, machine loading |
| Spherical (Polar) | RRP | Spherical - die casting, welding |
| SCARA | RRP (horizontal) | Circular - fast assembly of PCB, small parts |
| Articulated | RRR | Spherical large - welding, painting |

### Applications

- Welding, painting, pick and place, assembly, machine tending, palletising, material handling, hazardous work, packaging.

### Advantages / limits

- Fast, accurate, tireless, safe for humans, repeatable. Costly, limited sensing/thinking ability, need programming and safety fences.

## 6.5 Automatic Packaging Systems

- Packaging systems automatically fill, seal, label and pack products

at high speed with the help of sensors, PLCs and actuators. Used in food, pharma, beverages, cement, FMCG.

### Operations

- **Feeding / conveying -> Filling (volume or weight) -> Sealing / capping -> Labelling and date printing -> Inspection (weight, leak, vision) -> Carton packing -> Palletising.**

### Elements

- **Sensors:** photoelectric (bottle presence), proximity, load cell (weight), level sensor, colour/vision sensor.
- **Actuators:** conveyor motors, pneumatic cylinders (pushers, sealers), solenoid valves, servo for cutters.
- **Controller:** PLC with HMI (touch screen), counters, timers.




Fig: Automatic packaging line

- **Advantages:** high speed, uniform quantity, hygiene, less labour, less wastage, accurate count and weight.

## 6.6 Automatic Inspection Systems

- An automatic inspection system measures or checks the quality of

every product online (100% inspection) without human involvement and rejects defective parts.

## Methods / devices

- **Machine vision:** camera + lighting + image processing software to check dimensions, shape, colour, surface defects, presence of parts, bar codes.
- **Laser scanners / laser micrometers:** non contact measurement of diameter, thickness, profile.
- **Coordinate Measuring Machine (CMM):** touch trigger probe measures 3D coordinates.
- **Proximity / photoelectric and load cells:** presence and weight checks. Also ultrasonic and eddy current for hidden flaws.

## Working

- Part arrives on conveyor -> sensor triggers camera -> image captured -> processing and comparison with the standard -> **accept / reject decision** -> reject mechanism (air jet, pusher) removes the faulty part -> data stored for SPC (statistical process control).

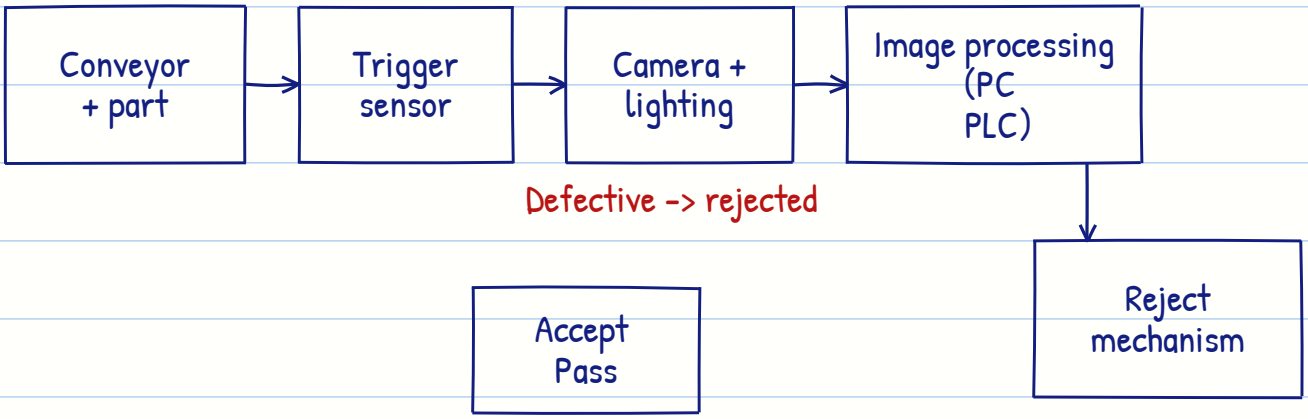



Fig: Automatic vision inspection system

- **Advantages:** 100% inspection, high speed, no fatigue or human error, consistent, data for quality improvement, non contact.

## Important Questions (Unit I)

- Define mechatronics. Explain the need and objectives of mechatronics in industry.
- Explain the mechatronics design process with a neat block diagram.
- Explain the key elements of a mechatronic system with an example.
- Describe a CNC machine with a block diagram; write its advantages.
- Write a short note on tool monitoring systems, FMS, industrial robots, automatic packaging and inspection systems.